

Chapter 1: Background Perspective

- 1.1. Impacts and Consequences of Technology on Society
- 1.2. Effects of Major Technological Developments such as Printing, Gunpowder, Mechanization, Computer, Organic Chemistry, Communication Satellites,
- 1.3. Cultural Motivations and Limitations,
- 1.4. Eastern vs. Western Philosophy of Change and Development,
- 1.5. Political and Limitations, Individual Freedoms versus Social Goals,
- 1.6. Exponential Growth. Alternative use of Scarce Resources,
- 1.7. Cause of International Tensions,
- 1.8. Risk and Overall Cost/Benefit,
- 1.9. Rational Analysis in Engineering Decision Making,
- 1.10. Education and Training of Technologists, Scientists and Engineers.

1.0 Society and its Elements

Definition of Society:

Many definitions of society; the definitions keeps changing, since society is dynamic.

- ❖ Schaefer and Lamm: the largest form of human group, which consists of people who share common heritage and culture. (Richard T. Schaefer and Robert P. Lamm)
- ❖ Ian Robertson: society gives content, direction and meaning to our lives, and we, in turn, in countless ways, reshape the society that we leave to the next generation. Society is a population that occupies the same territory, is subject to the same political authority and participates in a common culture.
- ❖ P. Gisbert: a complicated network of social relationships by which every human being is interconnected with his fellowmen.
- ❖ MacIver and Page: a system of usages and procedures, authority and mutual aid, of many groupings and divisions, of controls of human behavior and of liberties.
- ❖ Ginsberg: a collection of individuals unified by certain relation or modes of behavior, which mark them off from others who do not enter into these relations or who differ from them in behavior
- ❖ A.W. Green: the targets group to which any individual belongs
- ❖ John F. Cuber: a group of people who have lived long enough to become organized and to consider themselves and considered as a unit more or less distinct from other human units.

1.0.1 A settlement with the following criteria is called a society

- Population: A society has population. It can be small or big.
- Common territory: The people of a society occupy a common territory.

- Government or political authority: The people of a society have a common government and political authority.
- Common culture and a sense of relationship/membership and committed to group.

1.0.2 Essential elements of a society

- People and plurality: society consists of people, of both sexes, of different physical features (age, skin color, eye color, hair color, height, weight, face type)
- Likeness and differences: Members of a society are alike in many aspects. This “alike” bonds them, and they seek understanding and cooperation from each other. The members of a society are different in many ways too, like culture (language, religion, art, music, dance, food, dress, values, belief system), profession, interest, opinion, views etc.
- Cooperation and interdependence: Members of a society cooperate with each other, and they depend on each other. When the level of mutual cooperation reduces, society tends to disintegrate.
- Stable and dynamic: Society is relatively stable; the norms, values, and culture are normally stable/. Yet, society is dynamic. Some elements of a society change slowly, and some change rapidly, depending on external and internal factors.

1.0.3 Types/Evolution of Society

- Tribal: hunting and gathering society
- Pastoral: domesticate animals
- Horticultural: domesticate plants
- Agricultural/feudal: cultivation of crops, animal energy, irrigation, saving of seed
- Industrial: mechanized production, mass production
- Post-industrial: information and service, generation of knowledge

1.0.4 Factors/Drivers/Causes of social change

- Physical environment/Contact with other societies: Physically easily accessible society changes rapidly than those located in remote (difficult to access) areas due to frequent contacts with members of different societies.
- Natural causes: Earthquake, landslide, flood, desertification, and tsunami disintegrate social fabric and changes society.
- Technological: Information, knowledge and skills: Rate of social change depends on access to information, ability to put together the information into knowledge, ability to convert the knowledge into skills
- Anthropogenic (Human) activities: International War, Civil War, displacement for “developmental” activities, industrial accidents, mass migration, education, economic opportunity

1.0.5 Classical Theories of social change:

- I. **Cyclical:** ups and downs, birth and death, Oswald Spengler: approximate 1000 year cycle. Critics of this theory cite examples that do not follow the 1000 year cycle.
- II. **Evolution:** changing with time, getting complex with time, society develops through time and change.
- III. **Functionalist:** changes as required to keep the whole society functioning, when particular part of a society changes; based on the assumption that society is a stable, orderly system; August Comte (1798-1857), Herbert Spencer, Emile Durkheim, and Talcott Parsons propagate this theory. Society is composed of interrelated parts, each of which serves a function and contributes to the overall stability of the society. This theory emphasizes on changing role of different parts of a society to maintain stability of a society.
- IV. **Conflict:** Karl Marx (19th century) social class conflict between haves and have-nots. Expanded version of conflict theory finds social conflict between any groups in which the potential for inequity exists: racial, gender, religions, political, economic, and so on. Unequal groups usually have conflicting values and agendas, causing them to compete against one another. This constant competition between groups forms the basis for the ever-changing nature of society.

The triad is usually described in the following way:

- The thesis is an intellectual proposition.
- The antithesis is simply the negation of the thesis, a reaction to the proposition.
- The synthesis solves the conflict between the thesis and antithesis by reconciling their common truths and forming a new thesis, starting the process over.

Cyclical: The Greeks were the first to utilize this model. Plato spoke of eras of time when, initially, hope blossoms only to deteriorate as that era disintegrates. Spengler's view of change was very similar to that of the Greeks, but his model was dressed in a biological rather than uuologica1 garb. Culture, according to him, is the living entity of people, and culture is housed in the civilization of that it arises, develops, ripens, decays and falls never to return. An element of hope, however, was predicted in a similar model by Toynbee, who believed the cyclical change could be interrupted by the creative minority. More recently Sorokin considered that social changes follow a trendless cyclic pattern, i.e., like a swinging pendulum, culture moves in one direction and then back in another. The functionalist perspective, also called functionalism, is one of the major theoretical perspectives in sociology. It has its origins in the works of Emile Durkheim, who was especially interested in how social order is possible or how society remains relatively stable.

Functionalism interprets each part of society in terms of how it contributes to the stability of the whole society. Society is more than the sum of its parts; rather, each part of society is functional for the stability of the whole society. The different parts are primarily the institutions of society, each of which is organized to fill different

needs and each of which has particular consequences for the form and shape of society. The parts all depend on each other.

For example, the government, or state, provides education for the children of the family, which in turn pays taxes on which the state depends to keep itself running. The family is dependent upon the school to help children grow up to have good jobs so that they can raise and support their own families. In the process, the children become law-abiding, taxpaying citizens, who in turn support the state. If all goes well, the parts of society produce order, stability, and productivity. If all does not go well, the parts of society then must adapt to recapture a new order, stability, and productivity.

Functionalism emphasizes the consensus and order that exist in society, focusing on social stability and shared public values. From this perspective, disorganization in the system, such as deviant behavior, leads to change because societal components must adjust to achieve stability. When one part of the system is not working or is dysfunctional, it affects all other parts and creates social problems, which leads to social change.

The functionalist perspective achieved its greatest popularity among American sociologists in the 1940s and 1950s. While European functionalists originally focused on explaining the inner workings of social order, American functionalists focused on discovering the functions of human behavior. Among these American functionalist sociologists is Robert K. Merton, who divided human functions into two types: manifest functions, which are intentional and obvious, and latent functions, which are unintentional and not obvious. The manifest function of attending a church or synagogue, for instance, is to worship as part of a religious community, but its latent function may be to help members learn to discern personal from institutional values. With common sense, manifest functions become easily apparent. Yet this is not necessarily the case for latent functions, which often demand a sociological approach to be revealed.

Functionalism has received criticism for neglecting the negative functions of an event such as divorce. Critics also claim that the perspective justifies the status quo and complacency on the part of society's members.

Functionalism does not encourage people to take an active role in changing their social environment, even when such change may benefit them. Instead, functionalism sees active social change as undesirable because the various parts of society will compensate naturally for any problems that may arise.

Conflict theory emphasizes the role of coercion and power in producing social order. This perspective is derived from the works of Karl Marx, who saw society as fragmented into groups that compete for social and economic resources. Social order is maintained by domination, with power in the hands of those with the greatest political, economic, and social resources. When consensus exists, it is attributable to people being united around common interests, often in opposition to other groups.

According to conflict theory, inequality exists because those in control of a disproportionate share of society's resources actively defend their advantages. The masses are not bound to society by their shared values, but by

coercion at the hands of those in power. This perspective emphasizes social control, not consensus and conformity. Groups and individuals advance their own interests, struggling over control of societal resources. Those with the most resources exercise power over others with inequality and power struggles resulting. There is great attention paid to class, race, and gender in this perspective because they are seen as the grounds of the most pertinent and enduring struggles in society.

Whereas most other sociological theories focus on the positive aspects of society, conflict perspective focuses on the negative, conflicted, and ever-changing nature of society. Unlike functionalists who defend the status quo, avoid social change, and believe people cooperate to effect social order, conflict theorists challenge the status quo, encourage social change (even when this means social revolution), and believe rich and powerful people force social order on the poor and the weak. Conflict theorists, for example, may interpret an “elite” board of regents raising tuition to pay for esoteric new programs that raise the prestige of a local college as self-serving rather than as beneficial for students.

Whereas American sociologists in the 1940s and 1950s generally ignored the conflict perspective in favor of the functionalist, the tumultuous 1960s saw American sociologists gain considerable interest in conflict theory. They also expanded Marx's idea that the key conflict in society was strictly economic. Today, conflict theorists find social conflict between any groups in which the potential for inequality exists: racial, gender, religious, political, economic, and so on. Conflict theorists note that unequal groups usually have conflicting values and agendas, causing them to compete against one another. This constant competition between groups forms the basis for the ever-changing nature of society. Critics of the conflict perspective point to its overly negative view of society. The theory ultimately attributes humanitarian efforts, altruism, democracy, civil rights, and other positive aspects of society to capitalistic designs to control the masses, not to inherent interests in preserving society and social order.

There are many theories of social change. Generally, a theory of change should include elements such as structural aspects of change (like population shifts), Processes and mechanisms of social change, and directions of change.

- Hegelian: The classic Hegelian dialectic model of change is based on the interaction of opposing forces. Starting from a point of momentary stasis, Thesis countered by Antithesis first yields conflict, then it subsequently results in a new Synthesis.
- Marxist: Marxism presents a dialectical and materialist concept of history; Humankind's history is a fundamental struggle between social classes.
- Kuhnian: The philosopher of science, Thomas Kuhn argues in *The Structure of Scientific Revolutions* with respect to the Copernican Revolution that people are unlikely to jettison an unworkable paradigm,

despite many indications that the paradigm is not functioning properly, until a better paradigm can be presented.

- Heraclitan: The Greek philosopher Heraclitus used the metaphor of a river to speak of change thus, "On those stepping into rivers staying the same other and other waters flow" (DK22B12). What Heraclitus seems to be suggesting here, later interpretations notwithstanding, is that, in order for the river to remain the river, change must constantly be taking place. Thus one may think of the Heraclitan model as parallel to that of a living organism, which, in order to remain alive, must constantly be changing. A contemporary application of this approach is shown in the social change theory SEED-SCALE which builds off of the Complexity Theory subfield of Emergence.
- Daoist: The Chinese philosophical work Dao De Jing, I.8 and II.78 uses the metaphor of water as the ideal agent of change. Water, although soft and yielding, will eventually wear away stone. Change in this model is to be natural, harmonious and steady, albeit imperceptible.
- Resource-based economy: Jacque Fresco's concept of a resource-based economy that replaces the need for the current monetary economy, which is "scarcity-oriented" or "scarcity-based". Fresco argues that the world is rich in natural resources and energy and that — with modern technology and judicious efficiency — the needs of the global population can be met with abundance, while at the same time removing the current limitations of what is deemed possible due to notions of economic viability. Fresco's work in The Venus Project deals with physical reality and natural law rather than economic abstractions. The Venus Project's website says this: "The Venus Project is neither Utopian nor Orwellian, nor does it reflect the dreams of impractical idealists. Instead, it presents attainable goals requiring only the intelligent application of what we already know."[

1.0.6 Eastern and Western values of society

Eastern Values

Panchasheel:

1. Mutual respect for each other's territorial integrity and sovereignty,
2. Mutual non-aggression,
3. Mutual non-interference in each other's internal affairs,
4. Equality and mutual benefit, and
5. Peaceful co-existence
 - i. Social Harmony
 - ii. Sacrifice for the welfare of group
 - iii. Modesty and frugality
 - iv. Truth and Integrity

- v. Respect to elder, teacher, guest and ancestor
- vi. Maintaining culture and tradition
- vii. Helping the people in need (Paropakar)
- viii. Purity in thought and acts
- ix. Ritual as important as devotion or work

Western/American Values (Robin Williams, 1970)

- i. Achievement and success
- ii. Activity and work
- iii. Moral orientation
- iv. Humanitarianism
- v. Efficiency and practicality
- vi. Progress
- vii. Material comfort
- viii. Equality
- ix. Freedom
- x. External conformity
- xi. Science and rationality
- xii. Nationalism-patriotism
- xiii. Democracy
- xiv. Individual personality
- xv. Group-superiority themes

1.0.7 Brief History of engineering practice in eastern and western societies

1.1 Impacts and Consequences of Technology on Society

1.1.1 Socio-cultural Impact

Impact on	Impact
Livelihood	Livelihood is diversifying, many types of livelihood options available
Specialization of Profession	Specialization in profession is increasing.
Health	Access to health services is increasing, specific medicines are available, diagnosis process improving, genetic engineering is curing previously incurable diseases, assistance to specific needs of disabled persons getting

	better through technically designed equipment, welfare of aged citizens getting better
Education	Access to education services getting better, distance learning is improving access to education and latest information, e-book, e-library, standardization of educational quality through use of ICT technology, on line evaluation, on line test possible,
Family	Family relation getting complex due to in-vitro fertilization, test-tube baby, cloning, Family size reducing due to easier access to contraception, micro-family getting possible, life span increasing, several generation in family, change in status of female and disabled
Living Standard	Living standard increasing, HDI increasing, poverty decreasing, employment opportunity increasing, price of goods decreasing
Language	Use of international language increasing, language use getting standardized through mass media, brail script helping blind, sign language use increasing for deaf
Social Norms and Values	Social norms and values increasingly being challenged and altered or replaced, social class disintegrating, new economic class emerging
Family tradition, culture, heritage	Family traditions increasingly being challenged and altered or replaced, heritage preservation getting better
Urbanization	Urbanization increasing, concentration of resources and resource users
Commercialized Recreation	Traditional dances, music, drama and games giving ways to movies, video games, theme parks, pay per view programs